

JANTZEN AND WEIGHT FILTRATIONS FOR $\mathfrak{sl}_2(\mathbb{C})$ AND $\mathfrak{sl}_3(\mathbb{C})$

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ABSTRACT. We provide explicit constructions of the Jantzen filtration for Verma modules in the category $\mathcal{O}$ for $\mathfrak{sl}_2(\mathbb{C})$ and $\mathfrak{sl}_3(\mathbb{C})$. We compare their filtrations with the weight filtration of the corresponding perverse sheaves.

1. INTRODUCTION

Consider the BGG category $\mathcal{O}$ associated to a semisimple complex Lie algebra $\mathfrak{g}$. The Jantzen filtration of a Verma module $\Delta(\lambda)$ in $\mathcal{O}$ with highest weight $\lambda \in \mathfrak{h}^*$ is a remarkable tool which describes the submodule structure of $\Delta(\lambda)$. In particular, the filtration is directly computable, making use of a contravariant form on $\Delta(\lambda)$. In contrast, other filtrations, like the composition series and radical/socle filtrations, are only abstractly defined and difficult to compute in practice.

Jantzen conjectured in [Jan79] that his filtration is compatible in a natural way with embeddings of Verma modules. The truth of this conjecture would imply the truth of the Kazhdan–Lusztig (KL) conjecture: the multiplicities of simples in the associated graded pieces of the Jantzen filtration correspond to the coefficients of certain KL polynomials, from where the KL conjecture can be recovered by evaluating the polynomial at 1.

The Jantzen conjecture was settled by Beilinson–Bernstein [BB93] using significant extensions of the geometric techniques used in the proof of the KL conjecture. In particular, they considered an extra structure on the related category of perverse sheaves, which allowed them to interpret the Jantzen filtration as a *weight filtration*. Their proof involved passing to another related category of D -modules under Beilinson–Bernstein localisation, from where the weight filtration is realised as a *monodromy filtration* on vanishing cycles.

In this report, we will mostly focus on the algebraic Jantzen filtration and provide some simple examples of weight filtrations on the corresponding perverse sheaves. Section 2 will introduce contravariant forms, which will be the main computational tool used to define Jantzen filtration. Section 3 will step through the construction of the filtration, illustrated with examples from $\mathfrak{sl}_2(\mathbb{C})$ and $\mathfrak{sl}_3(\mathbb{C})$. Lastly, we provide the sheaf-theoretic versions of these examples in Section 4, giving only the necessary definitions to introduce the weight filtration.

Notation. Fix a triangular decomposition $\mathfrak{g} = \mathfrak{n}^- \oplus \mathfrak{h} \oplus \mathfrak{n}^+ = \mathfrak{n}^- \oplus \mathfrak{b}$. We denote by $U(\mathfrak{g})$ the universal enveloping algebra of $\mathfrak{g}$. The Verma module of highest weight $\lambda \in \mathfrak{h}^*$ is denoted by $\Delta(\lambda)$, and its maximal submodule and simple quotient $N(\lambda)$ and $L(\lambda)$ respectively. The projective cover of $L(\lambda)$ will be denoted by $P(\lambda)$. The Jordan–Hölder multiplicity of L in M is denoted by $[M : L]$, and the multiplicity of $\Delta(\lambda)$ in the standard filtration of M is denoted by $(M : \Delta(\lambda))$. Let Φ denote the

set of all roots, Φ^+ only the positive roots, and ρ be the half-sum of positive roots. Denote by Γ the set of $\mathbb{Z}_{\geq 0}$ -linear combinations of simple roots. The α -root space will be denoted by $\mathfrak{g}_\alpha$. The reflections s_α generate the Weyl group W of $\mathfrak{g}$, where α is a simple root, $w \cdot \lambda$ will denote the dot-action of w on a weight $\lambda \in \mathfrak{h}^*$, and $w_\circ$ the longest word of W . For $\mathfrak{sl}_2(\mathbb{C})$, there is only one simple reflection corresponding to the simple root α , which we will denote by s . The standard basis for $\mathfrak{sl}_2(\mathbb{C})$ is given by $\{e, h, f\}$. For $\mathfrak{sl}_3(\mathbb{C})$, there are two simple roots α and β , and we denote their corresponding reflections by s and t respectively. The standard basis for $\mathfrak{sl}_3(\mathbb{C})$ is given by $\{e_\alpha, e_\beta, e_{\alpha+\beta}, h_\alpha, h_\beta, f_\alpha, f_\beta, f_{\alpha+\beta}\}$. Lastly, $[M]$ will denote the symbol for $M \in \mathcal{O}$ in the Grothendieck group of $\mathcal{O}$.

2. CONTRAVARIANT FORMS

In this section, we provide the definition of a contravariant form, and a simple calculation. Let τ be the *Chevalley anti-involution*, which has the property that $\tau(\mathfrak{h}) = \mathfrak{h}$, $\tau(\mathfrak{g}_\alpha) = \mathfrak{g}_{-\alpha}$, $\tau(\mathfrak{g}_{-\alpha}) = \mathfrak{g}_\alpha$.

Definition 2.1. Let M be a $U(\mathfrak{g})$ -module. A symmetric bilinear form $\langle \cdot, \cdot \rangle_M : M \times M \rightarrow \mathbb{C}$ on M is called *contravariant* if

$$\langle u \cdot x, y \rangle_M = \langle x, \tau(u) \cdot y \rangle_M.$$

When M is a highest weight module, there is an explicit formula for the contravariant form on M . Let $\varepsilon^+ : U(\mathfrak{n}^+) \rightarrow \mathbb{C}$ be the map sending all non-constant PBW basis elements to 0, and similarly for ε^- . By the PBW theorem, we can consider the map $\pi := \varepsilon^- \otimes \text{id} \otimes \varepsilon^+ : U(\mathfrak{n}^-) \otimes U(\mathfrak{h}) \otimes U(\mathfrak{n}^+) \rightarrow U(\mathfrak{h})$. We have the following theorem.

Theorem 2.2 ([Sha73], [Hum08, Theorem 3.15]). *If M is a highest weight module of weight λ and v^+ a highest weight vector, then there is a contravariant form $\langle \cdot, \cdot \rangle_M$ on M uniquely determined by the value $\langle v^+, v^+ \rangle_M$ (up to scalar multiples), given by*

$$\langle u \cdot v^+, v \cdot v^+ \rangle_M = \lambda \circ \pi(\tau(u)v).$$

Furthermore, the form is non-degenerate if and only if $M \simeq L(\lambda)$.

Example 2.3. On $\Delta(\lambda)$ for $\mathfrak{sl}_2(\mathbb{C})$, we have that

$$\langle f \cdot v^+, f \cdot v^+ \rangle_{\Delta(\lambda)} = \lambda \circ \pi(ef) = \lambda \circ \pi(h - fe) = \lambda(h).$$

3. THE JANTZEN FILTRATION

Jantzen's idea in defining his filtration on Verma modules is to study the form $\langle \cdot, \cdot \rangle_{\Delta(\lambda)}$ as a *family of forms* depending on λ . We will make this precise in Section 3.2. Let us state some properties of this filtration.

Theorem 3.1 ([Jan79]). *Let $\lambda \in \mathfrak{h}^*$. Then a Verma module $\Delta(\lambda)$ has a finite filtration*

$$\Delta(\lambda) = \Delta(\lambda)^0 \supset \Delta(\lambda)^1 \supset \Delta(\lambda)^2 \supset \cdots,$$

that is $\Delta(\lambda)^i = 0$ for $i \gg 0$, satisfying:

- (a) Each nonzero subquotient $\Delta(\lambda)^i / \Delta(\lambda)^{i+1}$ has a nondegenerate contravariant form;
- (b) $\Delta(\lambda)^1 \simeq N(\lambda)$;

(c) The filtration pieces satisfy the equation

$$\sum_{i \geq 0} [\Delta(\lambda)^i] = \sum_{\substack{\alpha \in \Phi^+ \\ s_\alpha \cdot \lambda < \lambda}} [\Delta(s_\alpha \cdot \lambda)],$$

which is known as Jantzen's sum formula.

For $\mathfrak{sl}_2(\mathbb{C})$, the Jantzen filtration of a Verma module is simple enough that it coincides with its composition series (see Example 3.6). However, we start to see differences from $\mathfrak{sl}_3(\mathbb{C})$. By proving that the Jantzen filtration on Verma modules corresponds to the weight filtration on perverse sheaves, Beilinson–Bernstein showed the following algebraic property.

Theorem 3.2 ([BB93]). *The Jantzen filtration corresponds to the radical filtration.*

Here, the *radical filtration* of M is a filtration

$$M = \text{Rad}^0 M \supset \text{Rad}^1 M \supset \dots$$

such that $\text{Rad}^i M / \text{Rad}^{i+1} M$ is a maximal semisimple quotient of $\text{Rad}^i M$. Jantzen therefore gave a computational method to calculate this filtration.

3.1. Example: $\mathfrak{sl}_3(\mathbb{C})$. Before we give a sketch proof of Theorem 3.1, let us apply the sum formula to find the Jantzen filtration of Verma modules for $\mathfrak{sl}_3(\mathbb{C})$ with highest weight linked to 0. We first state a proposition which will be useful for describing the structure of $\Delta(\lambda)$ in terms of its simple submodule.

Proposition 3.3. *Let λ be integral, regular, and antidominant. Then for any $w \cdot \lambda \in W \cdot \lambda$, $\Delta(\lambda) \simeq L(\lambda)$ is the unique simple submodule of $\Delta(w \cdot \lambda)$. Furthermore, $[\Delta(w \cdot \lambda) : L(\lambda)] = 1$.*

Proof. The first part follows from Verma's embedding theorem. For the second part, it suffices to prove by BGG reciprocity that $(P(\lambda) : \Delta(w \cdot \lambda)) \leq 1$. Since $-\rho$ is dominant, we have that $\Delta(-\rho)$ is projective. Also since $w_\circ \cdot \lambda$ is dominant, $w_\circ \cdot \lambda + \rho$ is dominant integral. Therefore $L(w_\circ \cdot \lambda + \rho)$ is finite-dimensional, and so $M := \Delta(-\rho) \otimes L(w_\circ \cdot \lambda + \rho)$ is projective. It is a fact that M has a standard filtration with $\Delta(w \cdot \lambda)$ appearing at most once (see [EMTW20, Proposition 15.4]) and $\Delta(\lambda)$ appearing only at the top of the standard filtration. Since $M \twoheadrightarrow \Delta(\lambda)$, we have $P(\lambda) \subseteq M$ because M is projective, so $P(\lambda)$ has a standard filtration containing $\Delta(w \cdot \lambda)$ appearing at most once, as desired. $\square$

Let us compute the filtrations.

$\Delta(sts \cdot 0)$: We know that $\Delta(sts \cdot 0) \simeq L(sts \cdot 0)$ as $sts \cdot 0$ is antidominant, so the Jantzen filtration for $\Delta(sts \cdot 0)$ is trivial.

$\Delta(st \cdot 0)$: We have that $[\Delta(ts \cdot 0) : L(ts \cdot 0)] = 1$, and by Proposition 3.3, $[\Delta(ts \cdot 0) : L(sts \cdot 0)] = 1$ since $ts \cdot 0 = s \cdot (sts \cdot 0)$. These are all the possible candidates by Verma's theorem, so we have that

$$[\Delta(ts \cdot 0)] = [L(sts \cdot 0)] + [L(ts \cdot 0)],$$

yielding the Jantzen filtration

$$\Delta(ts \cdot 0) \overset{L(ts \cdot 0)}{\supset} L(sts \cdot 0) \overset{L(sts \cdot 0)}{\supset} 0,$$

which is the same as the composition series for $\Delta(ts \cdot 0)$. We have a similar filtration for $\Delta(st \cdot 0)$.

$\Delta(s \cdot 0)$: By the sum formula, we have

$$\begin{aligned} \sum_{i>0} [\Delta(s \cdot 0)^i] &= [\Delta(ts \cdot 0)] + [\Delta(st \cdot 0)] \\ &= [L(st \cdot 0)] + [L(ts \cdot 0)] + 2[L(sts \cdot 0)]. \end{aligned}$$

Now $N(s \cdot 0)$ is the maximal submodule of $\Delta(s \cdot 0)$. Since $\Delta(s \cdot 0)^1 \simeq N(s \cdot 0)$ must contain both $\Delta(ts \cdot 0)$ and $\Delta(st \cdot 0)$, so

$$[\Delta(s \cdot 0)^1] = [N(s \cdot 0)] = [L(st \cdot 0)] + [L(ts \cdot 0)] + [L(sts \cdot 0)],$$

and $[\Delta(s \cdot 0)^2] = [L(sts \cdot 0)]$. Therefore, the Jantzen filtration for $\Delta(s \cdot 0)$ is

$$\Delta(s \cdot 0) \supset^{L(s \cdot 0)} N(s \cdot 0) \supset^{L(st \cdot 0) \oplus L(ts \cdot 0)} L(sts \cdot 0) \supset^{L(sts \cdot 0)} 0.$$

Again, $\Delta(t \cdot 0)$ has a similar Jantzen filtration.

$\Delta(0)$: We have

$$\begin{aligned} \sum_{i>0} [\Delta(0)^i] &= [\Delta(s \cdot 0)] + [\Delta(t \cdot 0)] + [\Delta(sts \cdot 0)] \\ &= [L(s \cdot 0)] + [L(t \cdot 0)] + 2[L(ts \cdot 0)] + 2[L(st \cdot 0)] + 3[L(sts \cdot 0)]. \end{aligned}$$

Since $[\Delta(0) : L(sts \cdot 0)] = 1$, we have that $\Delta(0)^3 = L(sts \cdot 0)$. An analysis similar to the previous case yields

$$\begin{aligned} [\Delta(0)^1] &= [L(s \cdot 0)] + [L(t \cdot 0)] + [L(ts \cdot 0)] + [L(st \cdot 0)] + [L(sts \cdot 0)], \\ [\Delta(0)^2] &= [L(ts \cdot 0)] + [L(st \cdot 0)] + [L(sts \cdot 0)], \end{aligned}$$

so we have the Jantzen filtration

$$\Delta(0) \supset^{L(0)} N(0) \supset^{L(s \cdot 0) \oplus L(t \cdot 0)} \Delta(ts \cdot 0) + \Delta(st \cdot 0) \supset^{L(ts \cdot 0) \oplus L(st \cdot 0)} L(sts \cdot 0) \supset^{L(sts \cdot 0)} 0.$$

3.2. Proof Sketch of Theorem 3.1. In this subsection, we provide a proof sketch of Theorem 3.1 following [Hum08, §5.6-5.7]. We begin with a lemma which part (a) of the theorem is a special case. From there, part (b) follows readily. We will comment on the proof of part (c) but omit details.

Lemma 3.4 (“Key Lemma” [Hum08, Lemma 5.6]). *Let A be a principal ideal domain and M be a free A -module of rank r with a non-degenerate symmetric bilinear form $(\cdot, \cdot)$ with non-zero determinant D . Suppose $p \in A$ is a prime element. Define a filtration $M = M(0) \supset M(1) \supset \dots$ by*

$$M(n) := \{m \in M : (m, M) \subset p^n A\},$$

and write

$$\overline{A} := A/pA, \quad \overline{M} := M/pM, \quad \overline{M(n)} := M(n)/pM(n).$$

Then

- (a) $v_p(D) = \sum_{n>0} \dim_{\overline{A}} \overline{M(n)}$, where $v_p(a) = n$ when $p^n \mid a$ but $p^{n+1} \nmid a$, and
- (b) for each n , the bilinear form $(\cdot, \cdot)_n := p^{-n}(\cdot, \cdot)$ on $M(n)$ induces a non-degenerate form on $\overline{M(n)}/\overline{M(n+1)}$.

See [Hum08, Lemma 5.6] for a proof. We will just give an example illustrating the constructions in this lemma.

Example 3.5. Let $A = \mathbb{C}[T]$ and $M = Ae_1 \oplus Ae_2$ be the rank 2 A -module with bilinear form given by the matrix

$$\begin{pmatrix} 2T & 0 \\ 0 & T^3 \end{pmatrix},$$

that is for $m = ae_1 + be_2$, we have $(m, e_1) = 2Ta$ and $(m, e_2) = T^3b$. It follows that

$$\begin{array}{ll} M(0) = Ae_1 \oplus Ae_2 & \overline{M(0)} = \mathbb{C}e_1 \oplus \mathbb{C}e_2 \\ M(1) = Ae_1 \oplus Ae_2 & \overline{M(1)} = \mathbb{C}e_1 \oplus \mathbb{C}e_2 \\ M(2) = TAe_1 \oplus Ae_2 & \overline{M(2)} = \mathbb{C}e_2 \\ M(3) = T^2Ae_1 \oplus Ae_2 & \overline{M(3)} = \mathbb{C}e_2 \\ M(4) = T^3Ae_1 \oplus TAe_2 & \overline{M(4)} = 0 \\ \vdots & \vdots \end{array}$$

We see that $v_p(D) = v_p(2T^4) = 4$, which matches $\sum_{n>0} \dim_{\overline{A}} \overline{M(n)}$ as $\overline{M(n)} = 0$ for $n \geq 4$. Finally, the rescaled forms $(\cdot, \cdot)_n := T^{-n}(\cdot, \cdot)$ induce the following non-degenerate forms:

- $\overline{M(1)}/\overline{M(2)}$ is spanned by e_2 with $(e_1, e_1)_1 = T^{-1}(2T) = 2 \neq 0$;
- $\overline{M(2)}/\overline{M(3)}$ is trivial;
- $\overline{M(3)}/\overline{M(4)}$ is spanned by e_2 with $(e_2, e_2)_3 = T^{-3}(T^3) = 1 \neq 0$.

As hinted at the start of this section, we will define the filtration pieces $\Delta(\lambda)^i$ by extending to a family of filtration pieces first. Let $A := \mathbb{C}[T]$ and $K := \mathbb{C}(T)$ be its field of fractions. Let $\mathfrak{g}_A := A \otimes_{\mathbb{C}} \mathfrak{g}$ and $\mathfrak{g}_K := K \otimes_{\mathbb{C}} \mathfrak{g}$. We will use Lemma 3.4 to construct the filtration in the deformed weight spaces of $\Delta(\lambda)$, and recover the desired filtration by setting $T = 0$.

Given a weight $\lambda \in \mathfrak{h}^*$, set $\lambda_T := \lambda + T\rho \in \mathfrak{h}_K^{*-1}$. Then $\Delta(\lambda_T)$ is simple since $\lambda_T = \lambda + T\rho \in \mathfrak{h}_K^*$ is dot-antidominant as $\langle (\lambda + T\rho) + \rho, \alpha^\vee \rangle \notin \mathbb{Z}$ for all $\alpha \in \Phi$. By Theorem 2.2, the contravariant form $\langle \cdot, \cdot \rangle_{\Delta(\lambda_T)}$ is non-degenerate. Like normal Verma modules, the weight spaces of $\Delta(\lambda_T)$ are

$$\Delta(\lambda_T)_{\lambda_T - \nu} \quad \text{for } \nu \in \Gamma.$$

From the natural A -form $U(\mathfrak{g}_A) \simeq A \otimes_{\mathbb{C}} U(\mathfrak{g})$ in $U(\mathfrak{g}_K) \simeq K \otimes_{\mathbb{C}} U(\mathfrak{g})$, we get an A -form $\Delta(\lambda_T)_A \simeq U(\mathfrak{g}_A) \otimes_{U(\mathfrak{b}_A)} A_{\lambda_T}$, where A_{λ_T} is the rank 1 A -module where $\mathfrak{b}_A$ acts by λ_T ². The weight spaces have an A -valued non-degenerate symmetric bilinear form restricted from the contravariant form on $\Delta(\lambda_T)$. Write $\Delta_{\lambda_T - \nu} := \Delta(\lambda_T)_{\lambda_T - \nu} \cap \Delta(\lambda_T)_A$. Defining

$$\Delta(\lambda_T)_A^i := \sum_{\nu \in \Gamma} \Delta_{\lambda_T - \nu}(i),$$

we see that this defines a decreasing filtration of $\Delta(\lambda_T)_A$ by $U(\mathfrak{g}_A)$ -modules.

¹Here, choosing ρ specifies a *deformation direction*. One may think of A as the ring of regular functions on $\mathbb{C}\rho \subset \mathfrak{h}^*$ which is adopted in Beilinson–Bernstein’s geometric proof of the Jantzen conjecture. Deformation in the direction of a non-dominant weight results in different filtrations [Wil16].

²This construction is also called a *deformed Verma module*.

Finally, to get the corresponding filtration for $\Delta(\lambda)$, let

$$\Delta(\lambda)^i := \Delta(\lambda_T)_A^i / T\Delta(\lambda_T)_A^i,$$

which is the act of “setting $T = 0$ ”. Now by Lemma 3.4, the bilinear form $T^{-i}(\cdot, \cdot)$ induces a non-degenerate contravariant form on the quotients $\Delta(\lambda)^i / \Delta(\lambda)^{i+1}$. This proves part (a) of Theorem 3.1. Example 3.5 generalises to show that the filtration of each weight space eventually ends in 0, and since only finitely many weights are linked to λ , it follows that $\Delta(\lambda)^i = 0$ for large enough i .

Part (b) follows since the quotient $\Delta(\lambda) / \Delta(\lambda)^1$ is a highest weight module with a non-degenerate contravariant form, hence simple by Theorem 2.2, so $\Delta(\lambda) / \Delta(\lambda)^1 \simeq N(\lambda)$.

Lastly, it is a straightforward calculation to show the sum formula using Shapovalov’s determinant formula [Sha73], which we omit. For a proof, see [Hum08, §5.7].

3.3. Example of Weight Space Construction. We provide explicit calculations for the construction of Jantzen filtration in the case of $\mathfrak{sl}_2(\mathbb{C})$ and $\mathfrak{sl}_3(\mathbb{C})$.

Example 3.6. For $\mathfrak{sl}_2(\mathbb{C})$ with simple roots $\{\alpha\}$, consider $\Delta(\lambda)$ where $\lambda(h) = 0$. We expect the Jantzen filtration of $\Delta(\lambda)$ to be the same as its composition series

$$\Delta(\lambda) \supset \Delta(-\alpha) \supset 0.$$

The 1-dimensional weight space $\Delta(\lambda)_\lambda$ should only occur in the degree 0 piece of the filtration, whereas $\Delta(\lambda)_{-\alpha}$ should appear in both degree 0 and 1.

Now $\lambda_T = T\rho$. The weight space Δ_{λ_T} is spanned by a highest weight vector v^+ , so its contravariant form is just multiplication by a complex number. Then

$$\Delta_{\lambda_T}(i) = \begin{cases} Av^+ & i = 0 \\ TA v^+ & i = 1 \\ T^2 A v^+ & i = 2 \\ \vdots & \end{cases} \implies \Delta(\lambda)_\lambda^i = \begin{cases} \mathbb{C}v^+ & i = 0 \\ 0 & i \geq 1. \end{cases}$$

The weight space $\Delta_{\lambda_T - \alpha}$ is spanned by $a := f \cdot v^+$, so the contravariant form can be determined by the calculation

$$\begin{aligned} \langle f \cdot v^+, f \cdot v^+ \rangle_{\Delta(\lambda_T)} &= \langle v^+, ef \cdot v^+ \rangle_{\Delta(\lambda_T)} \\ &= \langle v^+, h \cdot v^+ \rangle_{\Delta(\lambda_T)} + \langle v^+, fe \cdot v^+ \rangle_{\Delta(\lambda_T)} \\ &= T \langle v^+, v^+ \rangle_{\Delta(\lambda_T)}, \end{aligned}$$

where we have used the fact that $fe \cdot v^+ = 0$ as v^+ is a highest weight vector, and that $h \cdot v^+ = (\lambda + T\rho)(h)v^+ = Tv^+$. This is where the formula in Theorem 2.2 comes from: writing elements in their PBW basis paired with the contravariance property kills all terms not in $U(\mathfrak{h})$ when acting on a highest weight vector. Therefore,

$$\Delta_{\lambda_T - \alpha}(i) = \begin{cases} Aa & i = 0 \\ Aa & i = 1 \\ T Aa & i = 2 \\ T^2 Aa & i = 3 \\ \vdots & \end{cases} \implies \Delta(\lambda)_{-\alpha}^i = \begin{cases} \mathbb{C}a & i = 0 \\ \mathbb{C}a & i = 1 \\ 0 & i \geq 2. \end{cases}$$

This shows that $\Delta(\lambda)_\lambda$ only occurs in degree 0, but $\Delta(\lambda)_{-\alpha}$ occurs in both degree 0 and 1.

Example 3.7. For $\mathfrak{sl}_3(\mathbb{C})$ with simple roots $\{\alpha, \beta\}$, consider $\Delta(\lambda)$, where $\lambda = \varpi$ is the fundamental weight corresponding to α , that is $\langle \varpi, \alpha^\vee \rangle = 1$ and $\langle \varpi, \beta^\vee \rangle = 0$. In particular, ϖ is dot-dominant so we should see the same Jantzen filtration as in Section 3.1. Let us just determine the weight space $\Delta_{\lambda_T - (\alpha + \beta)}$ and see what role it plays in the Jantzen filtration.

Now $\Delta_{\lambda_T - (\alpha + \beta)}$ has a basis $a := f_\alpha f_\beta v^+$ and $b := f_{\alpha + \beta} v^+$. The contravariant form with respect to this basis is given by

$$\begin{pmatrix} T^2 & -T \\ -T & 2T - 1 \end{pmatrix},$$

which can be calculated using the bracket relations. Therefore,

$$\Delta_{\lambda_T - (\alpha + \beta)}(i) = \begin{cases} Aa \oplus Ab & i = 0 \\ Aa \oplus TAb & i = 1 \\ Aa \oplus T^2Ab & i = 2 \\ TAa \oplus T^3Ab & i = 3 \\ \vdots & \end{cases} \implies \Delta(\lambda)_{\lambda - (\alpha + \beta)}^i = \begin{cases} \mathbb{C}a \oplus \mathbb{C}b & i = 0 \\ \mathbb{C}a & i = 1, 2 \\ 0 & i \geq 3. \end{cases}$$

Compare this calculation with the Jantzen filtration for $\Delta(0)$ in Section 3.1. Since $\Delta(\lambda)^3 \simeq \Delta(sts \cdot \lambda)$ has highest weight $sts \cdot \lambda \leq \lambda - (\alpha + \beta)$, we expect the weight space $\Delta(\lambda)_{\lambda - (\alpha + \beta)}$ to vanish for $i \geq 3$, which is confirmed above.

4. THE WEIGHT FILTRATION ON PERVERSE SHEAVES

In this section, we will provide sheaf-theoretic versions of the algebraic examples we have seen thus far for $\mathfrak{sl}_2(\mathbb{C})$ and $\mathfrak{sl}_3(\mathbb{C})$. We will black-box definitions and results, and not state them in full generality for the sake of exposition. For example, to state the definition of purity in Section 4.1, we would have to develop the theory of ℓ -adic sheaves. For a compact treatment of this section, see [Ach21, §5].

4.1. Weights and Purity. We will not go into the details of the weight filtration, but rather just give a quick (and slightly imprecise) introduction to see some examples.

For the flag variety G/B defined over $\overline{\mathbb{F}}_q$, where $\mathbb{F}_q$ is a finite field of characteristic p , there is a *Frobenius operator* Fr acting on its points. A sheaf $\mathcal{F}$ is said to be *pointwise pure of weight* w if the induced action of Fr on the stalk $\mathcal{F}_x$ at an $\mathbb{F}_{q^n}$ -point $x \in G/B$ has eigenvalues $(q^n)^{w/2}$ for every $n \geq 1$. We say that $\mathcal{F}$ is *mixed* if $\mathcal{F}$ has a finite filtration with pointwise pure quotients. A perverse sheaf has *weight* $\leq w$ if each cohomology sheaf $H^i(\mathcal{F})$ is mixed and has pointwise weights $\leq w + i$ (and *weight* $\geq w$ if $\mathbb{D}\mathcal{F}$ has weights $\leq -w$). Then $\mathcal{F}$ has *pure of weight* w if it has weights $\leq w$ and $\geq w$.

The main takeaway is that the weight w defines a grading on $\mathcal{F}$. We will black-box the details but illustrate this with two examples.

Example 4.1. If X is smooth of dimension d , then the *Tate sheaf* $\underline{\mathbb{C}}_X(\frac{1}{2})$ is pure of weight -1 . More generally, for any integer n , the sheaf $\underline{\mathbb{C}}_X(\frac{n}{2})$ is pure of weight $-n$. Note that this choice of normalisation is non-standard but convenient for our purposes.

The following theorem tells us that the intermediate extension behaves well with mixed sheaves of pure weight.

Theorem 4.2. *Let $h: X_0 \rightarrow X$ be a locally closed embedding. If $\mathcal{F}$ is mixed perverse of pure weight w on X_0 , then $h_{!*}\mathcal{F}$ is also mixed perverse of pure weight w on X . Moreover, every simple mixed perverse sheaf is pure.*

In particular, we have the following important example.

Example 4.3. Let X be an irreducible variety and $h: X_0 \rightarrow X$ be the inclusion of its smooth locus. Then the intersection cohomology complexes of X

$$\mathrm{IC}_X := h_{!*}\mathbb{C}_{X_0}[\dim X]$$

is of pure weight $\dim X$.

We come to the existence of a structure on mixed perverse sheaves called the *weight filtration*.

Theorem 4.4. *Every mixed perverse sheaf $\mathcal{F}$ has a canonical decreasing filtration $W_\bullet$ such that $W_i\mathcal{F}/W_{i+1}\mathcal{F}$ is pure of weight $-i$.*

In the remainder of the report, we will just sketch examples of weight filtrations for the flag variety of SL_2 and SL_3 .

4.2. Example: $G = SL_2$. Let us start with the simplest example in the case of $G = SL_2$. Its flag variety is $G/B = \mathbb{P}^1$ with Schubert stratification $\mathbb{P}^1 = \mathbb{A}^1 \sqcup \{pt\}$. Let $j: \mathbb{A}^1 \rightarrow \mathbb{P}^1$ and $i: \{pt\} \rightarrow \mathbb{P}^1$ be the open and closed inclusions. Under the correspondences, the simple objects of $\mathcal{O}_0$, namely $L(0)$ and $L(s \cdot 0)$, correspond respectively to the intersection cohomology sheaves

$$\mathrm{IC}_s := j_{!*}\mathbb{C}_{\mathbb{A}^1}[1] = \mathbb{C}_{\mathbb{P}^1}[1] \quad \text{and} \quad \mathrm{IC}_e := i_{!*}\mathbb{C}_{pt} = i_*\mathbb{C}_{pt}.$$

The Verma modules correspond to the sheaves

$$\Delta(0) \rightsquigarrow j_{!*}\mathbb{C}_{\mathbb{A}^1}[1] \quad \text{and} \quad \Delta(s \cdot 0) \rightsquigarrow i_{!*}\mathbb{C}_{pt}.$$

To determine the weight filtration for $j_{!*}\mathbb{C}_{\mathbb{A}^1}[1]$, consider the following distinguished triangle

$$j_!j^!\mathbb{C}_{\mathbb{P}^1}[1](\tfrac{1}{2}) \longrightarrow \mathbb{C}_{\mathbb{P}^1}[1](\tfrac{1}{2}) \longrightarrow i_*i^*\mathbb{C}_{\mathbb{P}^1}[1](\tfrac{1}{2}) \xrightarrow{+1},$$

which simplifies to

$$j_{!*}\mathbb{C}_{\mathbb{A}^1}[1](\tfrac{1}{2}) \longrightarrow \mathbb{C}_{\mathbb{P}^1}[1](\tfrac{1}{2}) \longrightarrow i_*\mathbb{C}_{pt}[1](\tfrac{1}{2}) \xrightarrow{+1}.$$

Now the last term is not in the perverse heart since $\dim pt = 0$, rotate the triangle to the left once. Then taking perverse cohomology yields the short exact sequence

$$0 \longrightarrow i_*\mathbb{C}_{pt}(\tfrac{1}{2}) \longrightarrow j_{!*}\mathbb{C}_{\mathbb{A}^1}[1](\tfrac{1}{2}) \longrightarrow \mathbb{C}_{\mathbb{P}^1}[1](\tfrac{1}{2}) \longrightarrow 0,$$

and gives the filtration

$$j_{!*}\mathbb{C}_{\mathbb{A}^1}[1](\tfrac{1}{2}) \xrightarrow{\mathbb{C}_{\mathbb{P}^1}[1](\tfrac{1}{2})} i_*\mathbb{C}_{pt}(\tfrac{1}{2}) \xrightarrow{i_*\mathbb{C}_{pt}(\tfrac{1}{2})} 0.$$

This filtration is the weight filtration since $\mathbb{C}_{\mathbb{P}^1}[1](\tfrac{1}{2})$ has pure weight 0 and $i_*\mathbb{C}_{pt}(\tfrac{1}{2})$ is pure of weight -1 .

4.3. Example: $G = SL_3$. Let us find the weight filtration of the sheaf corresponding to the Verma module $\Delta(s \cdot 0)$ for $\mathfrak{sl}_3(\mathbb{C})$. Consider the piece

$$\overline{\mathbb{A}_{ts}^2} = \mathbb{A}_{ts}^2 \sqcup \mathbb{A}_s^1 \sqcup \mathbb{A}_t^1 \sqcup \{pt\}$$

of the Schubert stratification for the flag variety of SL_3 . Note that we can make the identification

$$\mathbb{P}_s^1 \vee_{pt} \mathbb{P}_t^1 \simeq \overline{\mathbb{A}_s^1 \sqcup \mathbb{A}_t^1 \sqcup \{pt\}}.$$

Let $j: \mathbb{A}_{ts}^2 \rightarrow \overline{\mathbb{A}_{ts}^2}$ be the open embedding, $i: \overline{\mathbb{A}_s^1 \sqcup \mathbb{A}_t^1 \sqcup \{pt\}} \rightarrow \overline{\mathbb{A}_{ts}^2}$ be the closed embedding, and $\pi: \mathbb{P}_s^1 \sqcup \mathbb{P}_t^1 \rightarrow \mathbb{P}_s^1 \vee_{pt} \mathbb{P}_t^1$ be the projection map.

The mixed perverse sheaf of pure weight 0 corresponding to $\Delta(s \cdot 0)$ is

$$\Delta(s \cdot 0) \rightsquigarrow j_! \mathbb{C}_{\overline{\mathbb{A}_{ts}^2}}[2](1).$$

Using a similar method as the previous example, consider the distinguished triangle

$$j_! j^! \mathbb{C}_{\overline{\mathbb{A}_{ts}^2}}[2](1) \longrightarrow \mathbb{C}_{\overline{\mathbb{A}_{ts}^2}}[2](1) \longrightarrow i_* i^* \mathbb{C}_{\overline{\mathbb{A}_{ts}^2}}[2](1) \xrightarrow{+1},$$

which simplifies to

$$j_! \mathbb{C}_{\overline{\mathbb{A}_{ts}^2}}[2](1) \longrightarrow \mathbb{C}_{\overline{\mathbb{A}_{ts}^2}}[2](1) \longrightarrow \mathbb{C}_{\mathbb{P}_s^1 \vee_{pt} \mathbb{P}_t^1}[2](1) \xrightarrow{+1}.$$

Again, we rotate the triangle once so we can take perverse cohomology since $\dim \mathbb{P}_s^1 \vee_{pt} \mathbb{P}_t^1 = 1$, which yields the short exact sequence

$$0 \longrightarrow \mathbb{C}_{\mathbb{P}_s^1 \vee_{pt} \mathbb{P}_t^1}1 \longrightarrow j_! \mathbb{C}_{\overline{\mathbb{A}_{ts}^2}}[2](1) \longrightarrow \mathbb{C}_{\overline{\mathbb{A}_{ts}^2}}[2](1) \longrightarrow 0.$$

There is a short exact sequence of sheaves

$$0 \longrightarrow \mathbb{C}_{\mathbb{P}_s^1 \vee_{pt} \mathbb{P}_t^1}1 \longrightarrow \pi_* (\mathbb{C}_{\mathbb{P}_s^1}1 \oplus \mathbb{C}_{\mathbb{P}_t^1}1) \longrightarrow i_{pt*} \mathbb{C}_{pt}1 \longrightarrow 0,$$

where $i_{pt}: \{pt\} \rightarrow \mathbb{P}_s^1 \vee_{pt} \mathbb{P}_t^1$ is the embedding of the point. The last term is not perverse, so we pass to the derived level, rotate the last term, and apply perverse cohomology to get the short exact sequence

$$0 \longrightarrow i_{pt*} \mathbb{C}_{pt}(1) \longrightarrow \mathbb{C}_{\mathbb{P}_s^1 \vee_{pt} \mathbb{P}_t^1}1 \longrightarrow \pi_* (\mathbb{C}_{\mathbb{P}_s^1}1 \oplus \mathbb{C}_{\mathbb{P}_t^1}1) \longrightarrow 0.$$

Now $i_{pt*} \mathbb{C}_{pt}(1)$ is simple (it is the intersection cohomology sheaf $\mathrm{IC}_e(1)$), and the term on the right is the direct sum of the pushforward of constant sheaves along π . In particular, $\pi_* (\mathbb{C}_{\mathbb{P}_s^1}1 \oplus \mathbb{C}_{\mathbb{P}_t^1}1)$ is not of weight 0 but of weight $\frac{1}{2}$ since $\mathbb{C}_{\mathbb{P}_s^1}[1](\frac{1}{2})$ is of weight 0 (and similarly for t). We therefore get the filtration

$$j_! \mathbb{C}_{\overline{\mathbb{A}_{ts}^2}}[2](1) \xrightarrow{\mathrm{IC}_{ts}} \mathbb{C}_{\mathbb{P}_s^1 \vee_{pt} \mathbb{P}_t^1}1 \xrightarrow{\mathrm{IC}_s(\frac{1}{2}) \oplus \mathrm{IC}_t(\frac{1}{2})} \mathrm{IC}_e(1) \xrightarrow{\mathrm{IC}_e(1)} 0,$$

where $W_i j_! \mathbb{C}_{\overline{\mathbb{A}_{ts}^2}}[2] / W_{i+1} j_! \mathbb{C}_{\overline{\mathbb{A}_{ts}^2}}[2]$ is pure of weight $-i$, and each associated graded piece is semisimple.

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